

Beyond Reported Scores: Constructing an ESG Misrepresentation Index for the Tobacco Sector

| Trinity College Dublin
| ESG Analytics
| MSc Business Analytics

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1. Introduction

Environmental, Social, and Governance (ESG) frameworks have become central instruments through which institutional investors, regulators, and rating agencies evaluate corporate sustainability. As ESG-linked assets under management surpassed USD 35 trillion globally by 2022 (Global Sustainable Investment Alliance, 2023), the methodological integrity of ESG measurement has assumed critical importance, particularly in controversial industries where the consequences of flawed ESG measurement are greatest and likely to be material.

British American Tobacco plc (BAT) presents these limits directly. As one of the world's largest tobacco corporations, BAT has repositioned itself around a harm reduction narrative, centred on its ***"A Better Tomorrow"*** strategy and its New Generation Products (NGPs), including the Vuse e-cigarette and glo heated tobacco device. The company's composite normalised ESG score rose from 74.5 in 2023 to 81.0 in 2025, suggesting material sustainability progress. Yet evidence from public health bodies and BAT's own financial disclosures reveals a more troubling picture.

This report investigates: ***"To what extent do BAT's ESG ratings reflect its true societal impact, given evidence of geographic and product-level harm substitution rather than genuine harm reduction?"*** The empirical validity of this analysis hinges on a fundamental distinction between substantive harm reduction, which necessitates a quantifiable decline in aggregate societal harm, and harm substitution, where externalities are merely transferred across demographic or geographic boundaries. To test the legitimacy of BAT's sustainability narrative, this study develops a rigorous evaluation framework, utilising AHP-derived weighting and the ESG Misrepresentation Index (EMI) alongside predictive analytics to expose the material gaps in reported performance.

2. Literature Review

2.1. Structural Limitations of ESG Metrics

ESG ratings are imperfect proxies for real world corporate impact. Berg, Koelbel, and Rigobon (2022) demonstrate substantial divergence across rating agencies, with correlations between major providers averaging 0.54 compared with 0.99 for credit ratings. This divergence reflects methodological choices about which harms are considered material. In tobacco, those choices are biased against capturing product-level harm. Galbreath (2013) argues that ESG frameworks were designed for industries where operational improvements translate into reduced harm. For industries producing inherently harmful products, this assumption does not hold. A tobacco company can decarbonise its operations and publish GRI-aligned reports, all rewarded by ESG frameworks, while its core product continues to cause harm at scale.

2.2. The Scope 3 Boundary Problem

A critical limitation of mainstream ESG reporting is the exclusion of Scope 3 health externalities. BAT's Scope 3 emissions stand at 5,882 ktCO₂e, approximately 19 times its Scope 1 and 2 combined, yet no SBTi commitment has been made as of 2025 (BAT Annual Report, 2025). The addiction and mortality burden from product consumption is entirely absent from GRI Standards and the SASB Tobacco Industry Standard. Chatterji and Toffel (2010) observe that firms can score well on ESG through process improvements while their fundamental impact remains unchanged. This study terms that gap the boundary problem: ESG frameworks exclude the primary mechanism of harm in tobacco, which is the product itself.

2.3. Rating Divergence and ESG Arbitrage

BAT's rating profile makes this weighting failure concrete. Excluded from FTSE4Good and rated High Risk by Sustainalytics (30.4), the company holds a BBB from MSCI, and a composite score of 81.0 across CDP, ISS ESG, and S&P CSA. Kotsantonis and Serafeim (2019) argue that such divergence enables ESG washing, the signalling of commitment without

substantive harm reduction. This study advances the concept of ESG arbitrage: using weighting gaps to achieve high reported scores while transferring rather than reducing harm. The response is to elevate Social harm to a 50% weight via AHP.

3. Data Section

3.1. Sources

- **ESG ratings:** MSCI, Sustainalytics, CDP, ISS ESG, and S&P CSA scores (2023-2025), forming the composite reported score used in the EMI analysis.
- **Revenue:** BAT Annual Reports (2018-2025), covering combustible vs NGP revenue and regional breakdowns (AME, APMEA, US) for the crossover and share shift models.
- **Smokeless volume:** BAT Sustainability Reports (2020-2025), providing non-combustible user figures (millions) as input to the Holt-Winters forecast.
- **Social harm:** ASH UK and CDC/BRFSS via Kaggle (2015-2024), capturing youth e-cigarette uptake rates for S1 AHP scoring.
- **Environmental:** Ocean Conservancy litter audits and FAO tobacco production data via Our World in Data, underpinning E1/E3 sub-criteria.
- **Governance:** BAT climate and lobbying disclosures cross-referenced against SBTi records and CDP D-rating evidence for G1-G3 scoring.
- **Geographical distribution:** worldwide tobacco production (tonnes) by country in 2022, from Our World in Data from UN FAO.

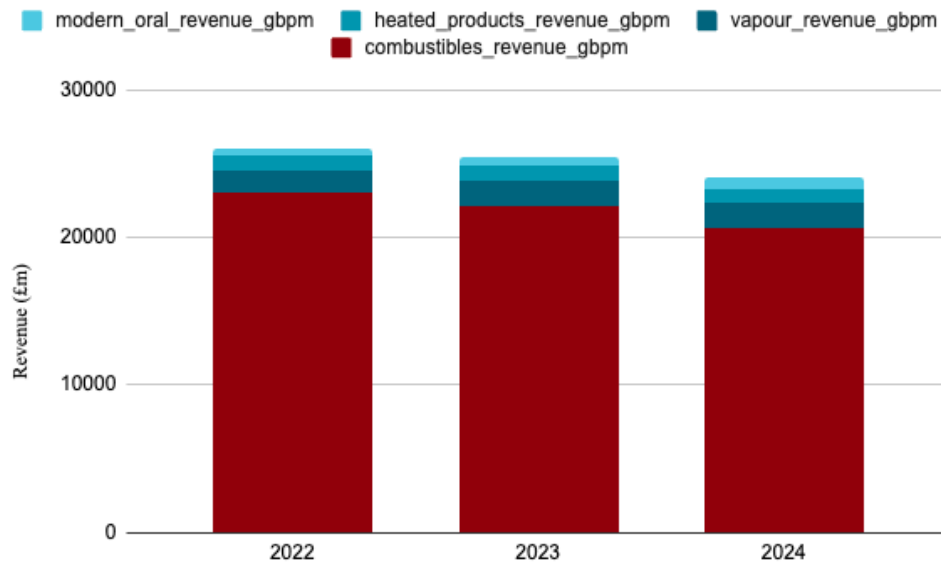


Fig. 1 BAT Revenue Split: Combustibles vs NGP (2022-2024)

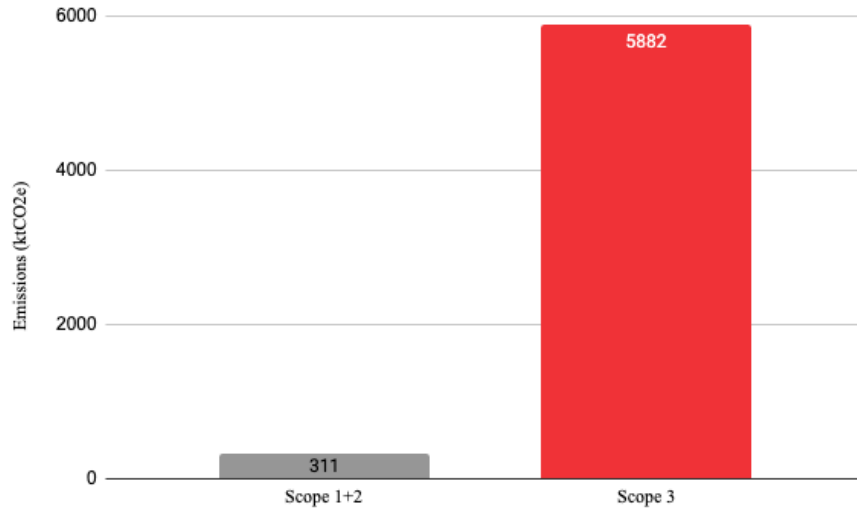


Fig.2 Scope 3 Emissions 19x Greater Than Scope 1+2 (2024)

3.2. Preprocessing

The ESG score is constructed by normalising multiple third-party ratings onto a consistent 0-1 scale and then averaging them. For example, MSCI ESG ratings (ranging from AAA to

CCC) are converted to a numerical scale out of seven, while Sustainalytics risk scores are inverted using a $(100 - \text{raw score})/100$ transformation to reflect lower risk as higher performance. CDP scores were normalised out of five, ISS ESG ratings were scaled relative to their six-point system, and S&P CSA percentages were used directly.

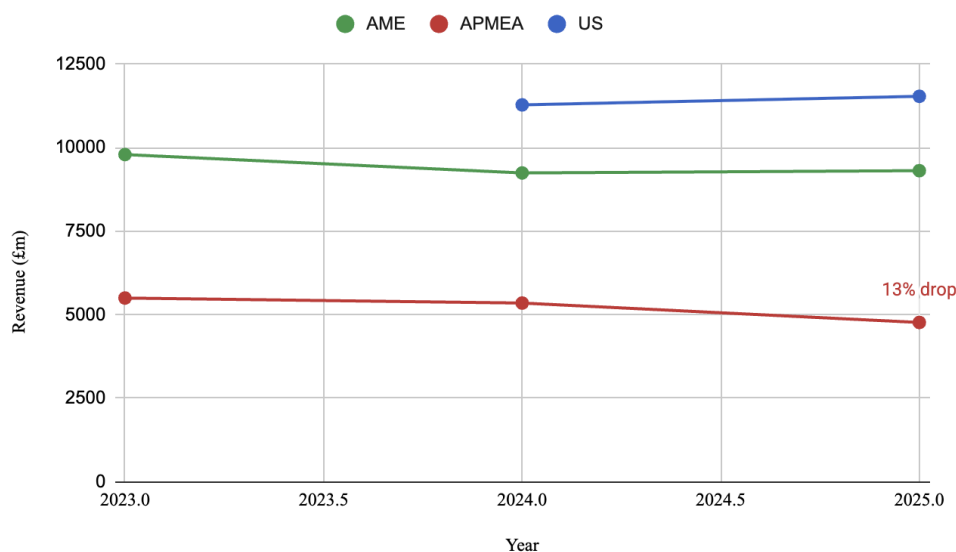


Fig. 3 APMEA Revenue Declining - US Resilient (2022-2039)

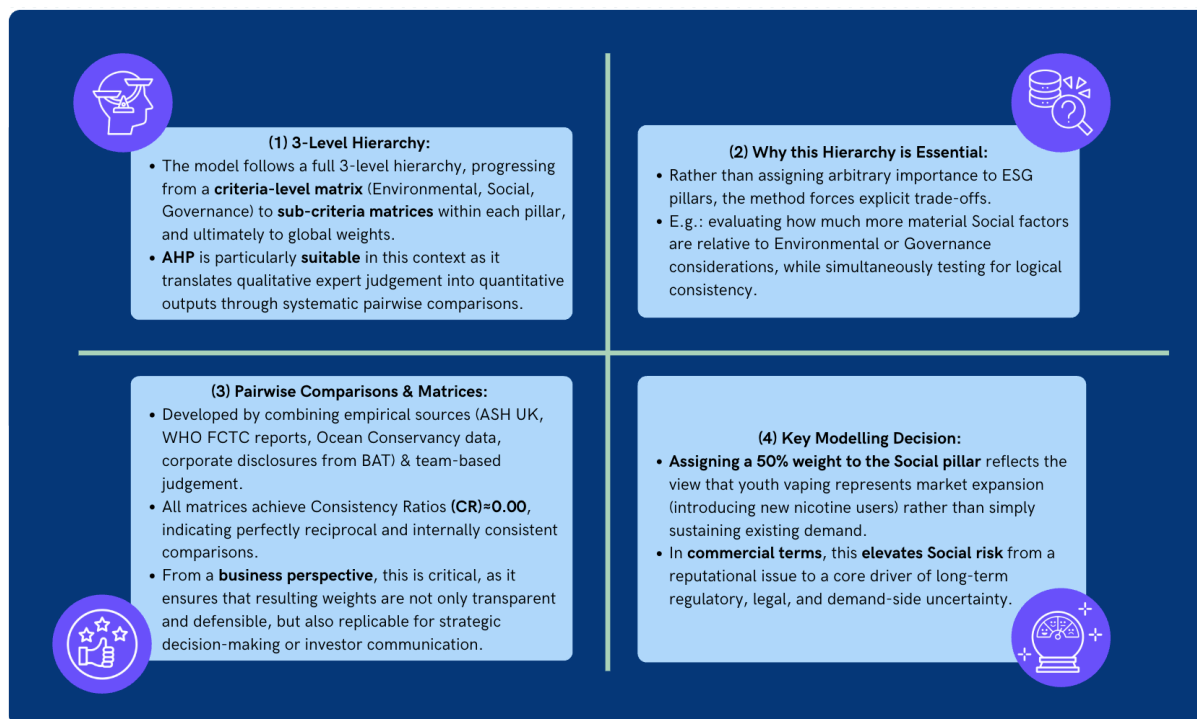
3.3. Limitations

BAT-sourced KPIs carry limited independent verification. All five rating providers use proprietary methodologies, meaning scores may reflect disclosure quality over true performance. Sub-Saharan Africa cannot be directly isolated from the APMEA segment in BAT's accounts.

4. Methodology

4.1. Analytic Hierarchy Process (AHP) ([Link to Dashboard](#))

This study applies the AHP to construct a structured, evidence-based ESG weighting framework.



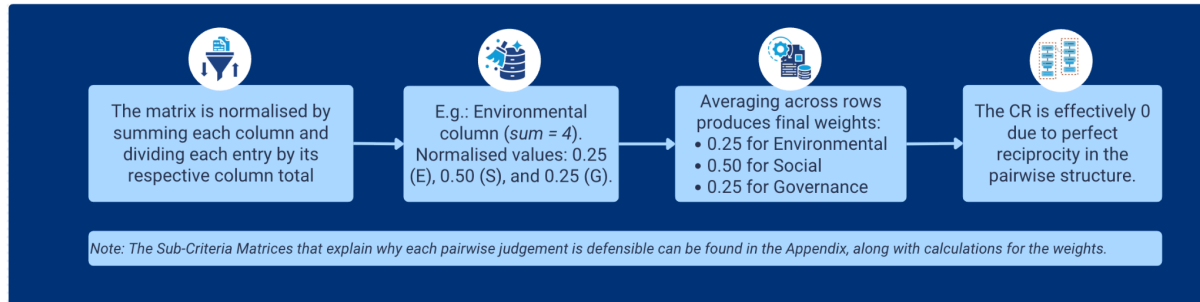
Diag.1 AHP Framework processes

4.1.1 Criteria-Level Matrix (E, S, G):

This matrix captures the relative importance of each ESG pillar. For instance, Social is judged to be twice as important as both Environmental and Governance, reflecting the direct and immediate impact of nicotine products on users and public health systems. Environmental impacts (waste and deforestation) are material but more indirect in their transmission to financial performance. Governance factors (disclosure practices and lobbying) are enabling mechanisms that shape the visibility and management of Environmental and Social risks rather than constituting primary impacts themselves.

	E	S	G
E	1	0.5	1
S	2	1	2
G	1	0.5	1

SUM	4	2	4
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Diag.2 Normalisation process, derived from converged sub-criteria matrices

4.1.2 Global Weights (Final Hierarchy):

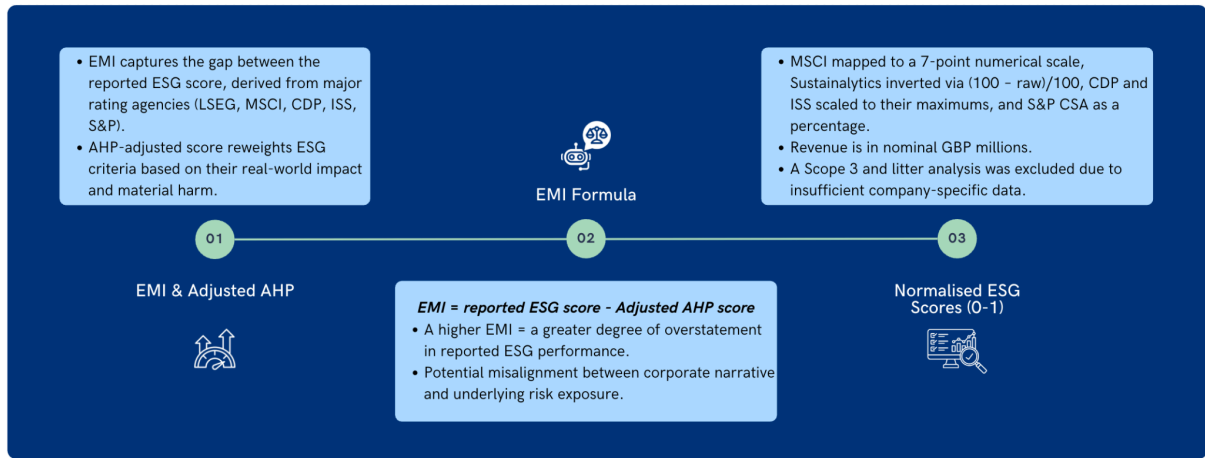
The Global Weights table is derived from the separated sub-criteria tables.

Criterion	Weight	Sub-Criterion	Local Weight	Global Weight
E	0.25	E1 butts	0.4	0.1
E	0.25	E2 vape waste	0.2	0.05
E	0.25	E3 deforestation	0.4	0.1
S	0.5	S1 youth vaping	0.5	0.25
S	0.5	S2 dependency	0.25	0.125
S	0.5	S3 SSA shift	0.25	0.125
G	0.25	G1 disclosure	0.4	0.1
G	0.25	G2 lobbying	0.3	0.075
G	0.25	G3 incentives	0.3	0.075
SUM	1			1

The final global weights combine pillar-level importance with sub-criteria prioritisation, producing a complete and operational ESG scoring framework. From a business standpoint, the results clearly concentrate risk in Social factors, especially youth vaping, which alone accounts for 25% of the total model weight. Environmental and Governance risks are more distributed but remain material, especially where they intersect with regulatory pressure, cost externalisation, and disclosure credibility. Overall, the model is fully replicable, internally consistent, and grounded in verifiable data sources, making it suitable for both academic analysis and strategic business applications.

4.2. Gap Analysis: ESG Misrepresentation Index (EMI)

The ESG Misrepresentation Index (EMI) is developed to quantify the divergence between externally reported ESG performance and an internally adjusted, materially grounded assessment.



Diag. 3 Explanations of EMI adjusted-AHP, and normalisation of ESG scores

Normalised Scores:

Metric	2023 raw	2024 raw	2025 raw	Normalisation Rule
MSCI ESG (AAA=7-CCC=1)	5	5	5	Out of 7 → 0.71

Sustainalytics (risk)	32.8	30.3	21.0	(100–raw)/100
CDP (Climate, Water, Forests avg)	4.0	5.0	5.0	Out of 5 → 0.8-1.0
ISS ESG (C=3, C+=3.5, B=4)	3.0	3.5	4.0	Out of 6 → 0.5-0.67
S&P CSA (0-100)	79	68	69	Out of 100 → 0.79, 0.68, 0.69

Aggregating these inputs produces a composite reported ESG score (scaled to 0-100) of 74.5 in 2023, 77.8 in 2024, and 81.0 in 2025. This upward trajectory reflects improvements captured by rating agencies, including CDP score upgrades and a reduction in Sustainalytics risk metrics. From a business perspective, this suggests strengthening ESG credentials in the eyes of external stakeholders and capital markets.

In contrast, the AHP-adjusted score is derived by applying the previously established ESG weights to sub-criteria scores grounded in external evidence. Each sub-criterion is assigned a score between 0 and 100 based on the severity and management of the associated risk.

Sub-Criterion Scores:

Sub-Criterion	Score (2025)	Evidence Anchor
E1 butts	45	Ocean Conservancy #1 pollutant, no BAT remediation target
E2 vape waste	40	No take-back scheme disclosed
E3 deforestation	55	Some disclosure, no SBTi for land-use

S1 youth vaping	20	ASH UK: tripled youth rates, severe penalty
S2 dependency	35	No net reduction in global nicotine users
S3 SSA shift	38	Expanding into lower-regulation markets
G1 disclosure	45	Scope 3 omission
G2 lobbying	55	D-rating on climate alignment
G3 incentives	58	Partial NGP linkage

Applying the AHP weights to these scores yields a 2025 adjusted ESG score of 39.0 (the complete calculation can be found in the Appendix).

EMI Calculation (Widening Gap):

Year	Reported ESG	AHP Score	EMI (Gap)	Interpretation
2023	74.5	35.0	39.5	Large initial gap
2024	77.8	37.0	40.8	Gap widens
2025	81.0	39.0	42.0	EMI up by 2.5 pts

Between 2023 and 2025, the reported ESG score increased by 6.5 points, while the AHP-adjusted score rose by only 4.0 points. This divergence indicates that ESG performance, as communicated to the market, is improving at a faster rate than underlying, materially weighted risk factors. In effect, BAT's ESG narrative is accelerating ahead of substantive impact mitigation.

This also suggests that reported ESG metrics are becoming progressively less reflective of material risk exposure. The primary driver of this divergence is the Social pillar, which carries the highest weight in the AHP model yet records consistently low performance

scores. Factors such as youth vaping, sustained dependency, and geographic risk transfer to lower-regulation markets exert a disproportionate negative impact on the adjusted score but are not equivalently penalised in mainstream ESG frameworks.

4.3. Forecast Analysis ([Link to Dashboard](#))

The “BAT Transition Dashboard” brings together time series forecasting (smokeless adoption and revenue evolution) and an optimisation layer to quantify how current trends translate into achievable 2030 financial and emissions outcomes.

1. Time Series Modelling

a. Holt-Winters Forecast for Smokeless Consumer Growth

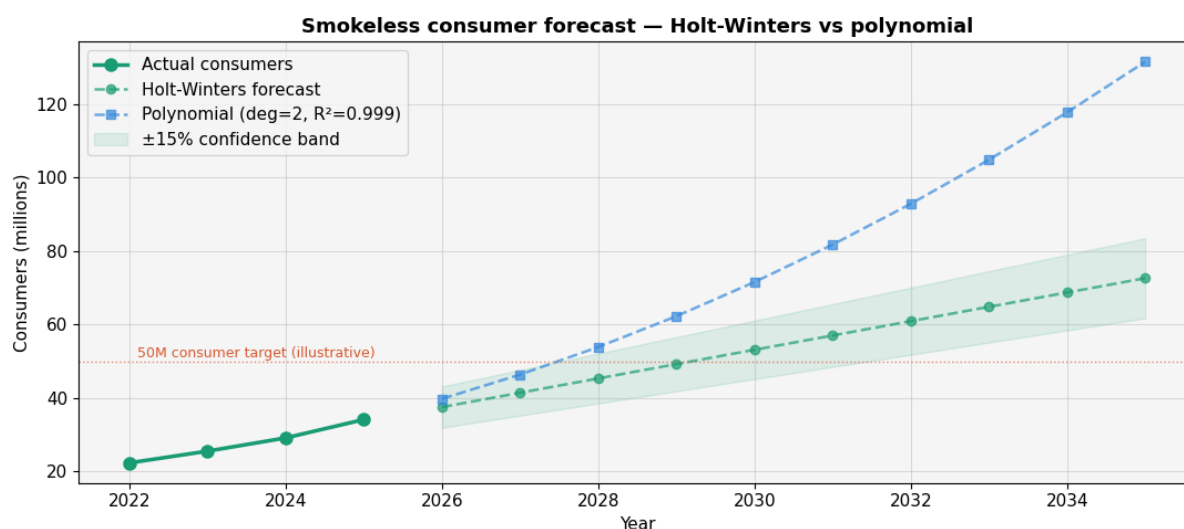
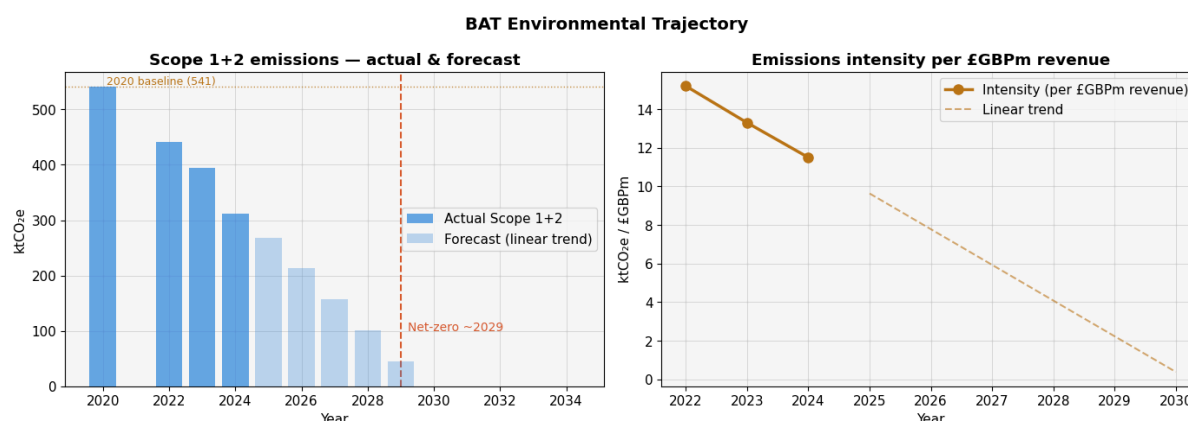


Fig.4 Smokeless Consumer Forecast (Holt-Winters vs. Polynomial)

A Holt-Winters exponential smoothing model projects steady growth in the smokeless consumer base, from 37.5 million in 2026 to 53.1 million by 2030 and 72.6 million by 2035 (CAGR ~5.6%). Holt-Winters is well-suited as it captures trend dynamics while weighting recent observations more heavily, making it appropriate for evolving adoption patterns.

Strategically, this growth is critical to offset declining combustibles, and the forecast provides a baseline for capacity planning and investment decisions, assuming current adoption drivers remain overall unchanged.

b. Emissions Time Series: Linear Trend and Net-Zero Projection



*Fig.5 BAT Environmental Trajectory: Scope 1&2 emissions and forecasts (L),
Emission Intensity per GPBm revenue (R)*

Linear regression shows a consistent decline in emissions (-55.7 ktCO₂e annually) with a strong fit ($R^2 = 0.9841$), implying potential net-zero by 2029, a notably rapid transition relative to industry averages. While this supports confidence in current decarbonisation efforts, the linear assumption may not hold as marginal abatement costs increase. Achieving the final reductions may therefore require additional measures such as technological upgrades or offsets.

4.3.1 Predictive Analytics

a. Revenue Crossover Model: When Does Smokeless Exceed Combustibles?

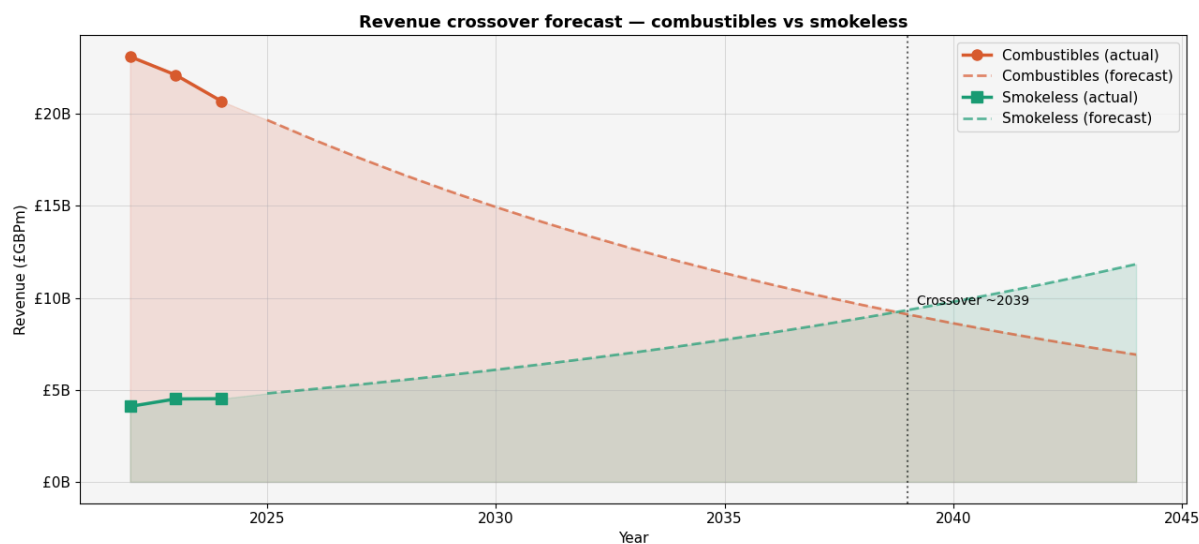


Fig.6 Revenue Crossover Forecast (Combustible vs Smokeless)

The model applies exponential decay to combustibles (-5.50% annually) and exponential growth to smokeless (+4.74%), reflecting typical transition dynamics. Revenues are projected to converge around 2039, when smokeless exceeds 50% of total revenue. This marks a structural inflection point, reinforcing the need for sustained investment in smokeless categories. Interestingly, the timing is sensitive to small changes in growth rates, which has implications for capital allocation and long-term planning.

b. Revenue Per Smokeless Consumer: Monetisation Efficiency

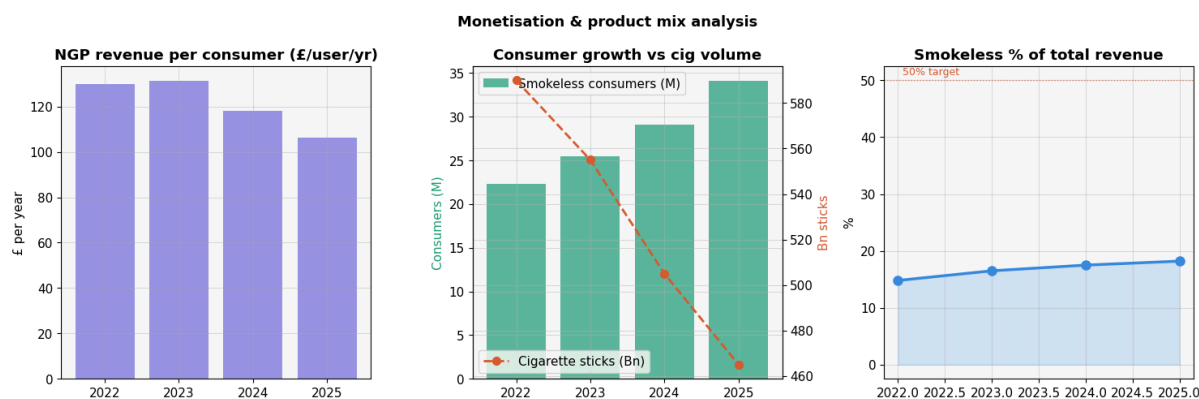


Fig.7 Monetisation & Product Mix Analysis: NGP Revenue per Consumer (L), Consumer Growth vs Cigarette Volume (C), Smokeless % of Total Revenue (R)

Revenue per user captures monetisation efficiency: stable or rising values indicate effective pricing and premiumisation, while declines may signal competitive pressure or market dilution. Combined with user growth, this metric is a key lever, signalling that improving monetisation can drive disproportionate revenue gains without equivalent increases in customer acquisition.

4.3.2 Data Mining

a. Geographic Revenue: Regional Share Shift Analysis

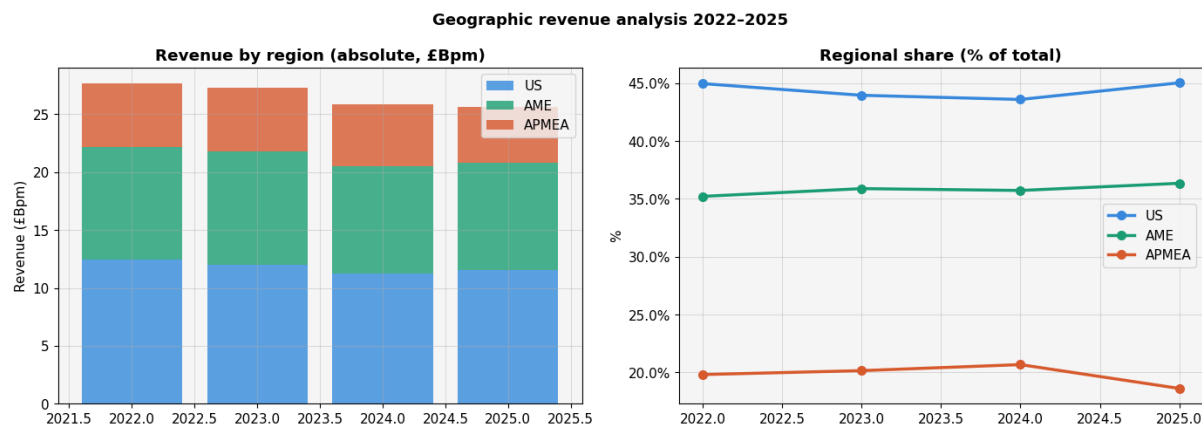


Fig.8 Geographic Revenue Analysis: Revenue by Region (L), Regional Share (R)

Regional trends are diverging. On one hand, AME (Africa and Middle East) shows moderate but volatile growth, APMEA (Asia-Pacific, Middle East, and Africa) deteriorates sharply (-10.86% by 2025), and on the other, the US shows early signs of recovery after contraction. This divergence supports a differentiated strategy that is, defending weaker regions while reallocating capital toward higher-growth or recovering markets.

b. Global Tobacco Production: Supply Concentration (FAO Data)

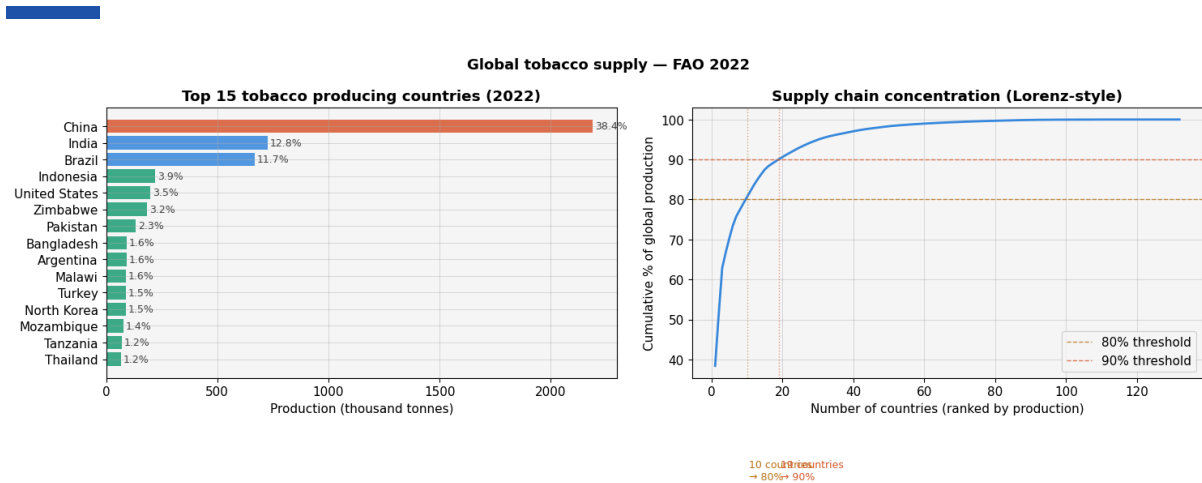


Fig.9 Global Tobacco Supply: Top 15 Tobacco Production Countries (L), Supply Chain Concentration (R)

Production is highly concentrated, having 5.70 million tonnes globally; China at 38.4%, with 10 countries accounting for 80%). This simplifies supplier management but increases exposure to geopolitical and climate risks. Diversification and stronger sourcing strategies could improve resilience.

4.3.3 Optimisation: Portfolio Transition Model

a. What Smokeless Growth Rate Is Needed To Offset Combustibles Decline?

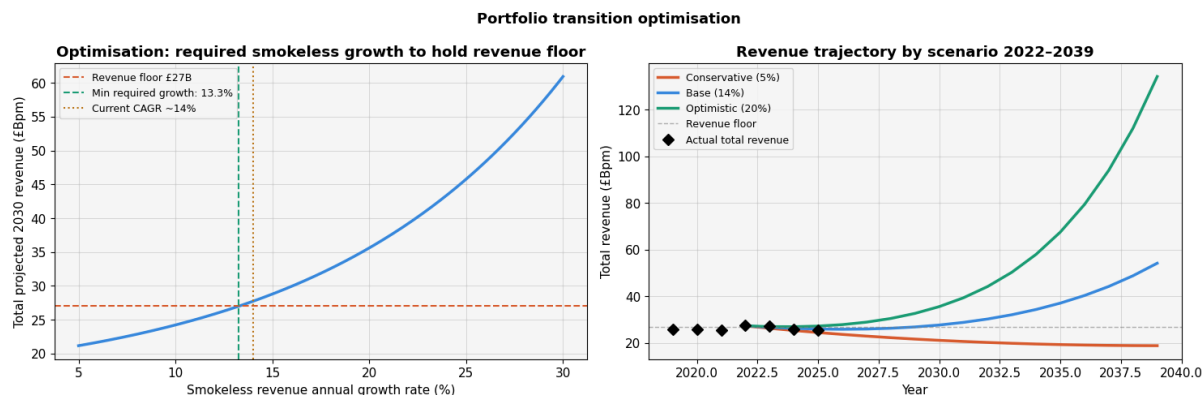


Fig.10 Portfolio Transition Optimisation: Optimisation (L), Revenue TRajjectory by Scenario (R)

This optimisation answers a key strategic question: what smokeless growth rate is needed to keep total revenue at £27 billion by 2030? The model calculates a minimum required CAGR of 13.3%, and the observed smokeless CAGR from 2022-2025 is approximately 14%, leaving a narrow margin of just 0.7 percentage points. This tight margin highlights the



importance of maintaining or accelerating current smokeless momentum to meet top-line targets.

5. Results

The AHP-EMI framework shifts the analysis from reporting ESG scores to materially weighted outcomes. The results are presented across 4 stages.

5.1. Why Mainstream Frameworks Overrate BAT

Fewer than 37% of LSEG's score comes from outcome and impact measures, with the remainder driven by policies and processes (Guerrero & Viteri, 2025). Materiality weights are derived from peer-group disclosure averages (LSEG, 2024), meaning that tobacco companies are benchmarked against each other rather than absolute harm. BAT's MSCI A rating against PMI's BBB, despite near identical portfolios, reflect superior disclosure, but not societal outcomes (Morningstar, 2021). The Sustainalytics improvement from 32.8 to 21.0 represents enhanced reporting rather than minimised societal harm (Sustainalytics, 2020). Importantly, the three dimensions that are most underweighted represent 47.5% of AHP weight.

5.2. How the AHP Framework Corrects This

The AHP model corrects the imbalance by explicitly prioritising Social factors and embedding product-related externalities into the weighing structure. When applied to the data analysed, this produces a materially lower adjusted ESG score than reported in ratings, widening the EMI gap over time. In fact, the BAT-adjusted AHP index for 2025 comes to 39.0, while the reported composite score is 81.0. The difference of 42 points makes the EMI figure, and this figure has been increasing annually to 39.5 points in 2023, 40.8 points in 2024, and 42.0 points in 2025. Importantly, this is not unique to BAT: peer companies converge under the same framework, reflecting a shared product-risk profile. The value of

AHP thus lies in exposing a systematic bias in mainstream ESG methodologies, namely the under-penalisation of product harm, as also shown in the sub-criterion scores: S1 = 20/100, S2 = 35/100, S3 = 38/100, and E1 = 45/100.

5.3. Does "Better Tomorrow" Hold Up?

The Royal College of Physicians (2025) estimates vapour products carry under 5% of combustible risk for adult users. However, it measures personal risk, not population harm. In England, 18-24 year-old vaping rose from 5.0% to 25.0% over 10 years. Among those who never smoked, the vaping rate within that age group increased to 34.3% (Jackson et al.,). US CDC data shows e-cigarette use at 11.0% among 18-24 year olds versus 4.5% across all adults (Cornelius et al., 2022). APMEA revenues fell 10.86% while AME revenues rose, confirming geographic harm substitution rather than net harm reduction.

5.4. Gap Analysis and Limitations

The ESG gap reflects structural limitations in conventional rating methodologies, which currently prioritise operational metrics (such as Scope 1 and 2 emissions reductions) over product-related externalities and Scope 3 effects. Additionally, Scope 3 emissions and downstream impacts are often insufficiently incorporated, further distorting the overall assessment. AHP introduces subjectivity, sensitivity analysis shows score variations of only ± 3 -5 points under different assumptions, without altering the direction of results. The EMI gap hence remains both significant and widening, supporting the case for sector-specific ESG approaches.

5.5. Summary of Findings of the Forecasts

Analysis	Method	Key finding
Consumer growth forecast	Holt-Winters time series	Smokeless consumers projected to reach ~65-75M by 2032 at the current trajectory

Emissions net-zero	Linear trend extrapolation	Scope 1+2 on track for near-zero ~2028-2030
Revenue crossover	Exponential decay/growth fit	Smokeless revenue likely to equal combustibles around 2034-2038
Monetisation efficiency	Predictive analytics	Revenue-per-user rising but still low (~£106/yr): key growth lever
Geographic shift	Data mining	APMEA losing share; US strengthening: regulatory divergence likely driver
Supply concentration	Lorenz analysis	80% of the global tobacco supply from just 5 countries, with China alone accounting for ~48%
Portfolio optimisation	Numerical optimisation	~8.5% smokeless CAGR is the minimum needed to hold £27B revenue floor by 2030

6. Discussion & Policy Implications

6.1. Main Findings

BAT's reported ESG scores are rising faster than any improvement in material harm outcomes, and the widening EMI captures that gap precisely. An AHP-adjusted score of 39.0 against a reported composite of 81.0 in 2025 is not measurement error. It reflects how ESG frameworks are built. Agencies score what is disclosed. A product whose primary harm occurs in the lungs of its users, years after purchase, falls outside the boundary of what most frameworks are designed to capture. This study terms the outcome ESG arbitrage: methodological gaps across rating systems allow a company to achieve strong reported scores while executing a strategy that transfers rather than reduces harm.

6.2. Limitations

AHP pairwise comparisons introduce anchoring bias. Adjusting the S1/S2 ratio from 2 to 3 shifts the final AHP score by approximately plus or minus 3 to 5 points, but does not close or reverse the EMI gap. BAT's revenue and emissions figures are largely self-reported, with KPMG assurance available only for selected years. Sub-Saharan Africa remains bundled within the APMEA segment, which limits the precision of the geographic harm analysis.

6.3. Policy Implications

Three groups face direct implications from these findings. Regulators should require independent verification of harm-reduction product claims before any ESG classification is granted, and WHO FCTC ratification gaps enabling a regulatory vacuum in Sub-Saharan Africa warrant multilateral attention. Investors should treat the divergence across FTSE4Good, Sustainalytics, and MSCI as a signal of greenwashing exposure rather than analytical disagreement. GRI and SASB should introduce a user-recruitment metric for tobacco that separates smoker conversion from market expansion, and move toward sector-specific harm-adjusted weighting for controversial industries.

7. Conclusion

This study set out to test whether BAT's ESG ratings reflect its real-world impact, and the data return a consistent answer. The ESG Misrepresentation Index widened from 39.5 to 42.0 between 2023 and 2025. The Social pillar, carrying 50% of the AHP weight, scored an average of 28 out of 100 on adjusted metrics. Youth vaping rates are rising, SSA combustible volumes are not falling, and nicotine dependency is not declining at a population level.

The forecast analysis sharpens this picture. Smokeless consumers are projected to reach 53.1 million by 2030, but if a significant portion of that growth is youth recruitment rather than smoker conversion, the headline figure measures market expansion rather than harm reduction. The observed smokeless CAGR sits only 0.7 percentage points above the



minimum required to hold the revenue floor, making the transition financially fragile as well as materially incomplete.

The primary contribution of this study is the framework itself.

The AHP-based EMI is replicable, grounded in verifiable sources, and applicable to other controversial industries. Its application to BAT exposes a structural problem in mainstream ESG assessment: frameworks reward good operational practice and disclosure quality, while a company's core product continues to cause harm at scale. Genuine progress in the tobacco sector requires dependency reduction, not rebranding, and the frameworks designed to measure that progress need to reflect where the harm actually falls.



8. Appendix

Methodology in Depth

Sub-Criteria Matrices (Why Each Pairwise Judgement is Defensible):

a. Environmental (E1, E2, E3)

	E1	E2	E3	Weight
E1	1	2	1	0.4
E2	0.5	1	0.5	0.2
E3	1	2	1	0.4

Where:

Weights (row averages):

- $E\ weight = (0.25 + 0.25 + 0.25) / 3 = 0.25$
- $S\ weight = (0.50 + 0.50 + 0.50) / 3 = 0.50$
- $G\ weight = (0.25 + 0.25 + 0.25) / 3 = 0.25$

Consistency ratio (CR) ≈ 0.00 because the matrix is perfectly reciprocal:

($S/E = 2$, $E/S = 0.5$; $S/G = 2$, $G/S = 0.5$; $E/G = 1$)

Within the Environmental pillar, cigarette butt pollution (E1) and tobacco-related deforestation (E3) are weighted equally (*value = 1*), each representing large-scale, visible, and insufficiently regulated externalities. Evidence from Ocean Conservancy (2023) identifies cigarette butts as the most prevalent form of global beach pollution, while FAO and WHO FCTC estimates attribute approximately 200,000 hectares of annual deforestation to tobacco production.

In contrast, vape-related waste (E2) is assigned a lower relative importance, with both E1 and E3 judged to be twice as significant (*value = 2*). While electronic waste from vaping

devices is a growing environmental concern, mainly given the presence of lithium batteries and non-recyclable components, it remains smaller in absolute volume compared to cigarette butt waste. Additionally, current corporate disclosures do not fully capture the lifecycle impact of vape products, suggesting an underrepresentation of this risk in reported data. As a result, E2 is assigned a weight of 0.20, reflecting a lower but still material contribution to overall environmental impact.

The matrix maintains a consistency ratio of approximately 0.00, as all pairwise comparisons are perfectly reciprocal (e.g., $E1/E2 = 2$ and $E2/E1 = 0.5$). This ensures that the weighting structure is logically coherent and methodologically robust, supporting its use in both academic analysis and business-oriented ESG evaluation.

b. Social (S1, S2, S3) - Most Important Pillar (weight 0.50)

	S1	S2	S3	Weight
S1	1	2	2	0.5
S2	0.5	1	1	0.25
S3	0.5	1	1	0.25

The Social pillar is assigned the highest overall weight (0.50), reflecting its direct linkage to core revenue generation, long-term demand sustainability, and regulatory exposure. Within this pillar, youth vaping (S1) is judged to be twice as significant as both dependency among existing users (S2) and geographic expansion into Sub-Saharan Africa (S3). This reflects the view that youth uptake constitutes net market expansion (introducing new nicotine users) rather than simply maintaining an existing consumer base. Empirical evidence supports this weighting: ASH UK (2024) reports a tripling of youth vaping rates between 2021 and 2024, while CDC (2023) data indicates that 7.7% of U.S. high school students vape. WHO

findings further show that youth usage rates exceed adult prevalence in several key markets, reinforcing the scale and immediacy of this risk.

A critical additional justification underpins the equivalence assigned to S2 and S3 (*value* = 1). Dependency (S2) perpetuates harm among existing users, sustaining long-term consumption and associated health burdens. In parallel, expansion into SSA (S3) replicates this harm profile in lower-regulation environments, effectively shifting rather than reducing global risk exposure. While the mechanisms differ (addiction retention vs geographic expansion), their overall impact is of similar magnitude. This interpretation is supported by BAT annual reports, which show revenue growth in SSA offsetting declining volumes in Western markets, indicating no net reduction in the global population of nicotine-dependent users.

From a business perspective, this dynamic is particularly significant: it suggests that risk is being redistributed rather than mitigated, with potential implications for future regulatory intervention, litigation exposure, and ESG investor scrutiny. Lastly, the matrix is fully consistent and strongly evidence-backed.

c. Governance (G1, G2, G3)

	G1	G2	G3	Weight
G1	1	1	2	0.4
G2	1	1	1	0.3
G3	0.5	1	1	0.3

The Governance pillar captures the extent to which corporate structures, disclosures, and incentives shape the management and transparency of ESG risks. Within this matrix, disclosure gaps (G1) are assigned a slightly higher importance relative to lobbying (G2) and

executive incentives (G3). This judgement is supported by evidence indicating that BAT's Scope 3 emissions (5,882 ktCO₂e) are approximately 19 times greater than its combined Scope 1 and 2 emissions, yet remain insufficiently addressed within externally validated frameworks. The absence of an SBTi commitment as of 2025 further reinforces this as a material omission. From a business standpoint, such gaps increase the likelihood of future regulatory intervention, investor pressure, and reputational risk.

A critical additional judgement is that lobbying (G2) and incentives (G3) are of equal importance (*value = 1*). Both factors influence how ESG risks are operationalised within the firm: lobbying shapes the external regulatory environment, while incentive structures determine internal strategic priorities. Evidence from InfluenceMap (2023) assigns BAT a "D" rating for alignment between its climate lobbying and stated commitments, suggesting potential inconsistencies in external positioning. At the same time, long-term incentive structures include targets related to next-generation products, but short-term incentives lack explicit metrics tied to harm reduction, indicating partial but incomplete alignment.

The matrix structure reflects these relationships, confirming that the set of judgements is logically coherent and methodologically robust.

AHP Calculation:

$$\begin{aligned} 2025 \text{ AHP} - \text{adjusted score} &= \\ (45 \times 0.10) &+ (40 \times 0.05) + (55 \times 0.10) + (20 \times 0.25) + (35 \times 0.125) + \\ + (45 \times 0.10) &+ (55 \times 0.075) + (58 \times 0.075) = \\ = 4.5 &+ 2.0 + 5.5 + 5.0 + 4.375 + 4.75 + 4.5 + 4.125 + 4.35 = \\ = 39.0 \end{aligned}$$

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